HALALNiaga

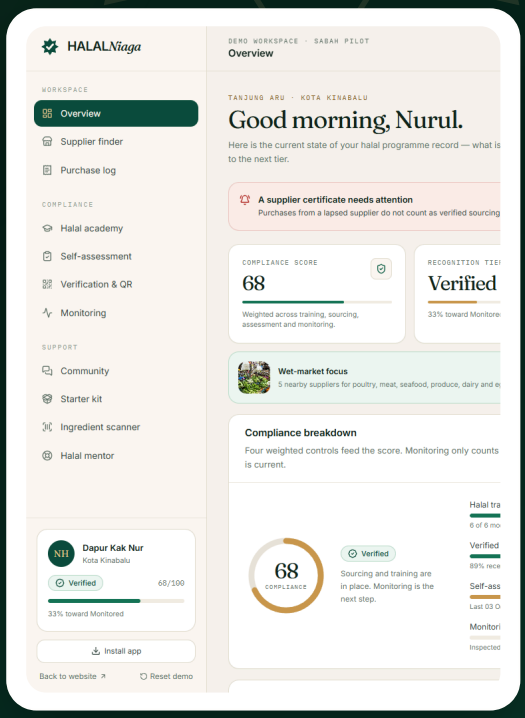
SYSTEM BRIEF · 2026

PWA PROTOTYPE · SABAH PILOT

A digital halal ecosystem for small food *vendors.*

Helping vendors source verified wet ingredients nearby, complete halal training, and publish a record customers can scan at the stall.

Wet ingredients first | Installable PWA | Works offline



Scan to open the live demo

halalniaga.pages.dev

01 · EXECUTIVE SUMMARY

One record that connects the stall to its *sources.*

HALALNiaga is a digital ecosystem for small food vendors: hawkers, stalls, food trucks, caterers and night-market traders. It brings together nearby certified or verified wet-ingredient suppliers, practical halal training, purchase records, self-assessment and programme recognition.

The prototype is built as a PWA: installable from the browser, usable offline after the first visit, and powered by local demo data. No backend, payment or real account is required.

06 Integrated modules

03 Recognition tiers

05 Wet-ingredient suppliers

01 Customer QR record

Client brief

- **Wet ingredients first.** Chicken, fish, vegetables, eggs and milk — the categories that are hardest to trace.
- **PWA, not an app store.** The prototype can be installed and used on a phone.
- **Limited budget.** Static hosting, local data and no server bill.

Find

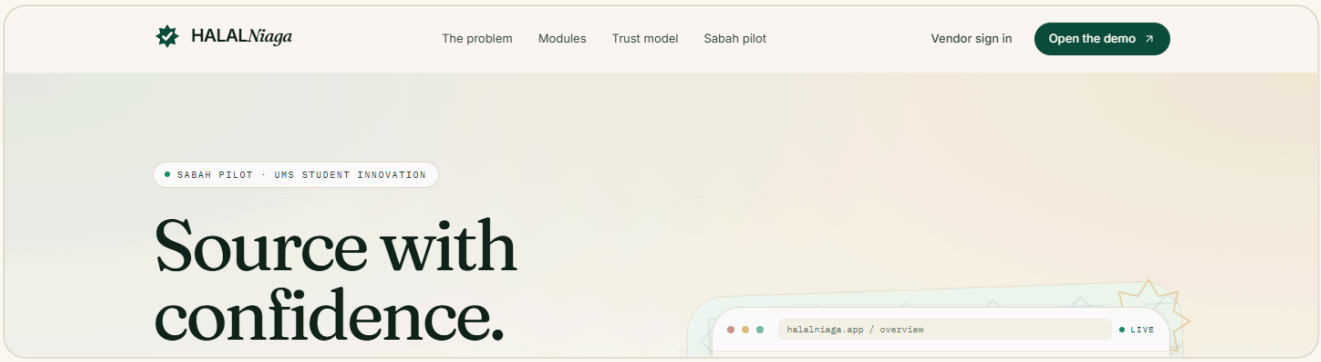
Nearby suppliers, product-level status and certificates on one screen.

Prove

Purchases, training and assessment become one evidence trail.

Publish

A stall QR shows the programme record a customer can check.



Landing page · core message, certification boundary and PWA proof.

02 · THE PROBLEM

The gap is not intention. It is *evidence.*

Small vendors want to do the right thing. What they lack is a practical way to prove what they already do — and a clear next step when something is missing.

01

Suppliers are hard to verify

Certificates sit in drawers or inboxes, rarely at the 6am wet-market decision about which chicken, fish or vegetable crate to buy.

02

Halal knowledge is uneven

Many still reduce halal to “no pork, no alcohol”, missing slaughter, storage and separation.

03

Certification feels out of reach

A small stall assumes the official process is too complex or expensive to even explore.

04

No purchase records

Without receipts or supplier references, nobody can prove where the chicken, fish or produce came from.

05

Cross-contamination is invisible

Shared boards, knives and oil are common — and almost never documented or corrected.

06

Wet ingredients move faster than paperwork

Chicken, fish, vegetables, eggs and milk are bought several times a week, often from different stalls. A record built for monthly invoices cannot keep up with a wet market.

> Do not turn a small vendor into an *administrator*. Build the record from the work they already do — buy, cook, store, clean, serve.

The design stance

03 · ECOSYSTEM

Three parties. One chain that can be *verified.*

HALALNiaga is not a supplier directory. It connects vendors, suppliers and programme guidance in one shared record.

01

Small vendor

Buys wet ingredients, cooks and serves. The system records evidence along the daily work.

02

Verified supplier

Publishes product-level halal status and reaches nearby small stalls.

03

Guidance & monitoring

Trains, advises, inspects and recognises along a clear ladder.

SOURCE
Verified supplier
Certificate and product status recorded.

→

OPERATION
Small vendor
Buy, record, cook and store correctly.

→

CONFIDENCE
Customer
Scans the QR and sees the programme record.

The programme supports this chain through training, self-assessment, inspection and recognition. HALALNiaga records and monitors — it does not issue a halal certificate.

Design principle

Compliance is assessed on evidence, not identity. Malaysian halal certification is open to Muslim and non-Muslim operators who meet the requirements. This record is designed to support compliance, not replace the official process.



Kota Kinabalu wet market · the daily source for small vendors. Photo: Bahnfrend, CC BY-SA 4.0.

04 · SIX MODULES

Six modules. One continuous *record.*

Training, sourcing, assessment and inspection stop living in separate notebooks. Every module feeds the same record.

01 SEARCH

Supplier Finder
Find nearby wet-ingredient suppliers; filter certificate and product status.

02 TRAINING

Halal Academy
Six short modules, 80% pass mark, programme completion certificate.

03 RECORD

Purchase Tracking
Every purchase becomes evidence: supplier, product, quantity and receipt.

04 CHECK

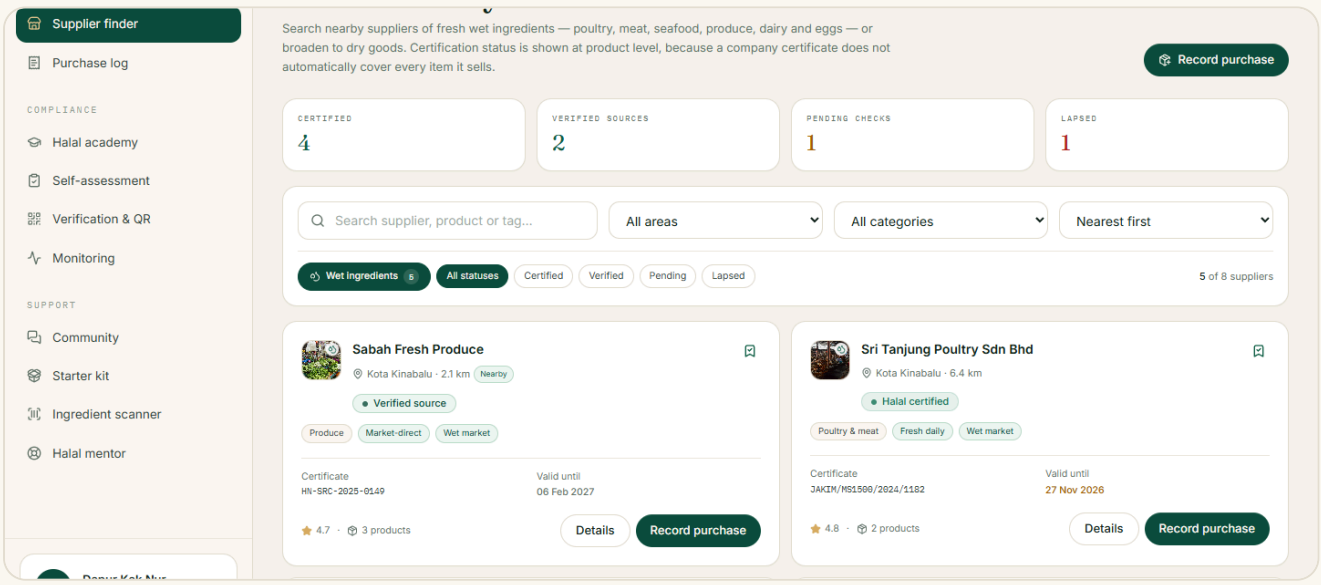
Self-Assessment
Eight weighted questions, corrective actions, valid for 90 days.

05 TRUST

QR Verification
A public profile for customers: training, sources, assessment and inspection.

06 RECOGNITION

Monitoring & Recognition
Three tiers, programme inspections and periodic re-assessment.



Supplier Finder · 5 wet-ingredient suppliers, sorted nearest first.

05 · VENDOR FLOW

From registration to the *QR at the stall.*

Six steps. Each step adds evidence to the same record — without adding another paper ritual.

01	**Register the stall** Name, location, menu and the ingredients that matter most.	Stall profile
02	**Halal training** Six short modules, then a scored quiz for each module.	Quiz score
03	**Find wet-ingredient suppliers** Filter wet categories, sort by distance, check product status.	Verified source
04	**Record purchases** Supplier, product, quantity, date and receipt number.	Source trail
05	**Self-assessment** Check eight controls; see the weak point and the correction.	Assessment score
06	**Recognition & QR** Move up through inspection; publish the customer QR.	Stall QR

What it produces

One record that answers: who supplied it, what was bought, when, and what evidence exists. The record can be reviewed at inspection.

Starter kit

Storage labels, equipment colour guide, hygiene checklist and QR sticker — all printable from the app.

`/app/starter-kit`



Dashboard · score, tier and alerts at a glance.

06 · CUSTOMER FLOW

One scan. The evidence behind the *food.*

Customers do not need to install an app or create an account. The QR sticker at the stall opens the programme record in the browser.

01
Scan
The phone camera opens the record.

→

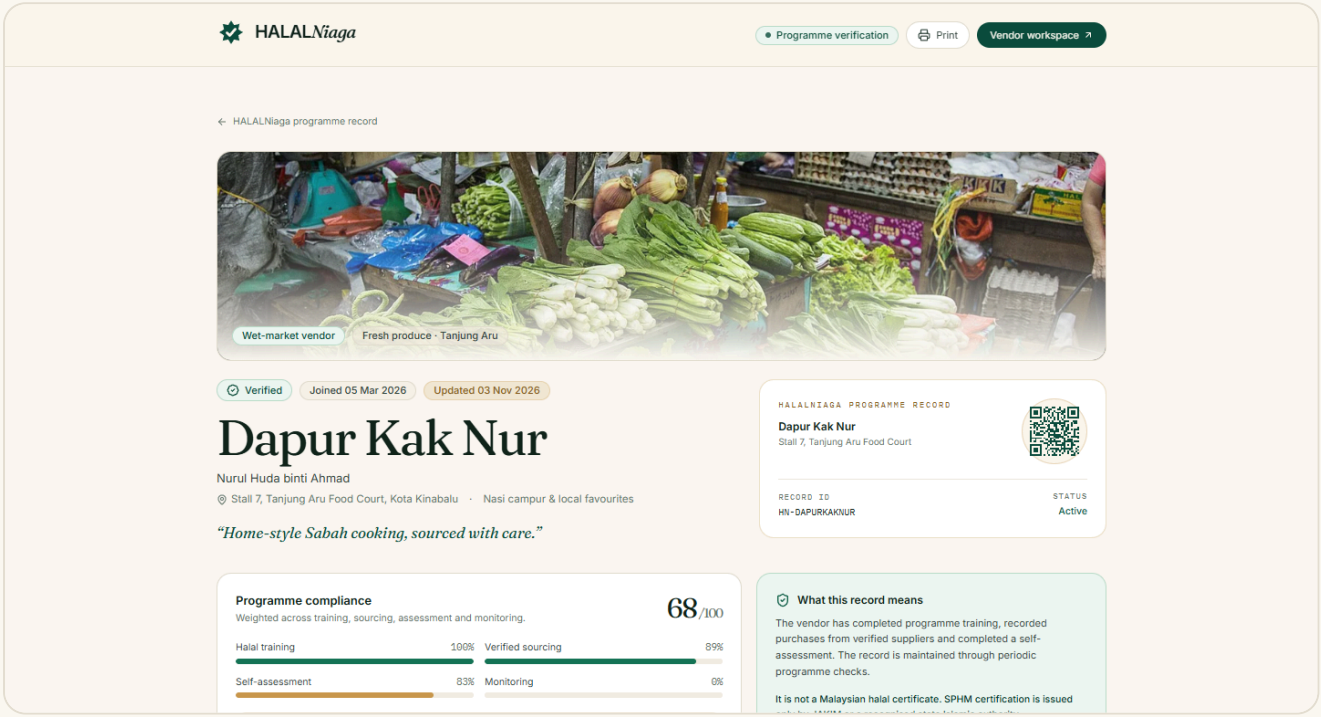
02
View the record
Tier, training, sources and last check date.

→

03
Check the evidence
Verified suppliers, assessment and inspection.

→

04
Decide with confidence
A decision based on information, not a logo alone.



Public record · /vendor/dapur-kak-nur · scannable without an account.

Not a JAKIM halal certificate
This record states programme recognition and the last check date. Official halal certification remains with JAKIM or the state Islamic authority.

07 · RECOGNITION MODEL

Recognition that has to be *maintained.*

Tiers are not permanent labels. The evidence behind them expires — and the record shows when it needs renewal.

01 Tier 1

Learner

All six modules passed at 80% or higher.

`Programme completion certificate`

02 Tier 2

Verified Participant

Learner + 80% of recent purchases from verified sources + self-assessment 80% within 90 days.

`Badge + stall QR`

03 Tier 3

Monitored Participant

Verified + programme inspection passed within 180 days + re-assessment in the current cycle.

`Monitoring badge + inspection date`

The important boundary

HALALNiaga does not issue halal certificates, does not use the Malaysian halal logo, and does not replace inspections by JAKIM or the state Islamic authority. Every QR record states this.

Programme recognition · Not SPHM · Not the JAKIM logo

Official reference: JAKIM Halal Malaysia · halal.gov.my

TIER	MAIN REQUIREMENT	RECOGNITION
Learner	6/6 modules passed · quiz ≥ 80%	Completion certificate
Verified	Learner + sources ≥ 80% + assessment ≥ 80%	Badge + QR
Monitored	Verified + inspection passed ≤ 180 days	Monitoring badge

ASSESSMENT
90 days
Refresh each cycle.

INSPECTION
180 days
Current programme inspection.

SUPPLIER CERTIFICATE
Monitored
Check expiry on every purchase.

08 · PWA, TECHNOLOGY & BUDGET

A PWA you install, not an *app store download.*

The brief asked for a PWA prototype on a limited budget. HALALNiaga installs from the browser, caches its shell, and keeps working when the signal drops at the market.

Install from the browser

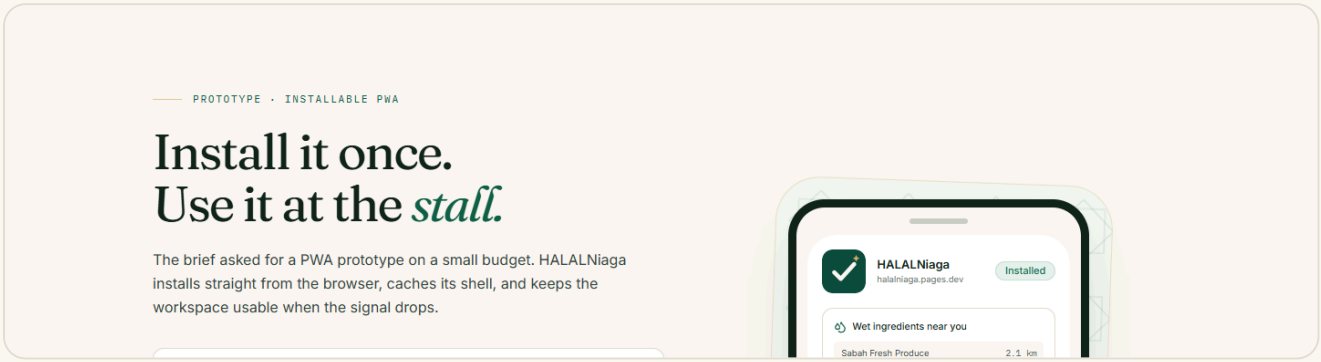
No app store, no review process. Add it to the home screen in seconds.

Offline-ready

The app shell and demo data remain after the first visit.

No server bill

Static hosting on Cloudflare Pages, local data, no backend or payments.



Landing page · “Install it once. Use it at the stall.”

Prototype architecture

CLIENT
PWA
React · Vite

→

CACHE
Service Worker
Workbox precache

→

DATA
localStorage
Local demo record

Hosting: Cloudflare Pages · static assets · SPA fallback.

Future: D1/Workers, auth, supplier onboarding, JAKIM directory integration.

Android

Menu ⋮ → Install app / Add to Home screen.

iPhone

Share → Add to Home Screen.

Desktop

Install icon in the address bar.

Launch: The installed app opens directly to the vendor workspace (`/app`), not the landing page.

Note: This prototype has no backend, payment, email or real supplier integration. Data is stored in the device browser.

09 · WET-INGREDIENT FOCUS

Wet ingredients come *first.*

Chicken, fish, vegetables, eggs and milk are bought several times a week and change suppliers. They are the hardest categories to trace — and the most important to monitor.

Poultry & meat
Chicken, beef, mutton
2 sources

Seafood
Fish, prawns, shellfish
1 source

Produce
Kangkung, tomato, shallots
1 source

Dairy & eggs
Eggs, fresh milk
1 source

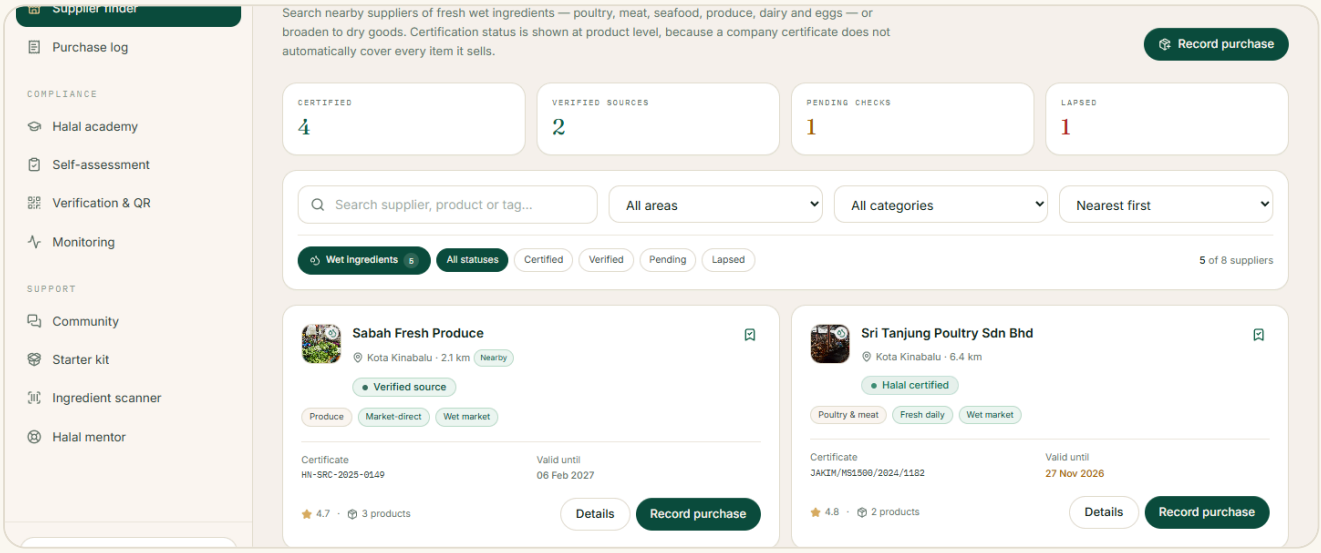
Why wet ingredients?

- Short buying cycle: stock changes daily, not monthly.
- Higher cold-chain and cross-contamination risk.
- Supplier certificates can expire or change scope.
- Customers are more likely to ask where the chicken, fish and vegetables came from.

In the app

The **Wet ingredients** button filters five wet-ingredient suppliers. The list is sorted **nearest first**, with distance and a "Nearby" marker for sources within 5 km.

`/app/suppliers?wet=1`



Wet-ingredient filter · 5 of 8 suppliers · nearest-first order.

10 · DEMO

Try it in *five minutes.*

The demo uses the sample vendor Dapur Kak Nur and pilot data. No real registration is required.

Direct links

Landing	`halalniaga.pages.dev`
Workspace	`/app`
Wet ingredients	`/app/suppliers?wet=1`
Public record	`/vendor/dapur-kak-nur`
Starter kit	`/app/starter-kit`

Scan to open the demo

`halalniaga.pages.dev`

Demo account

`nurul@dapurkaknur.my`

`demo1234`

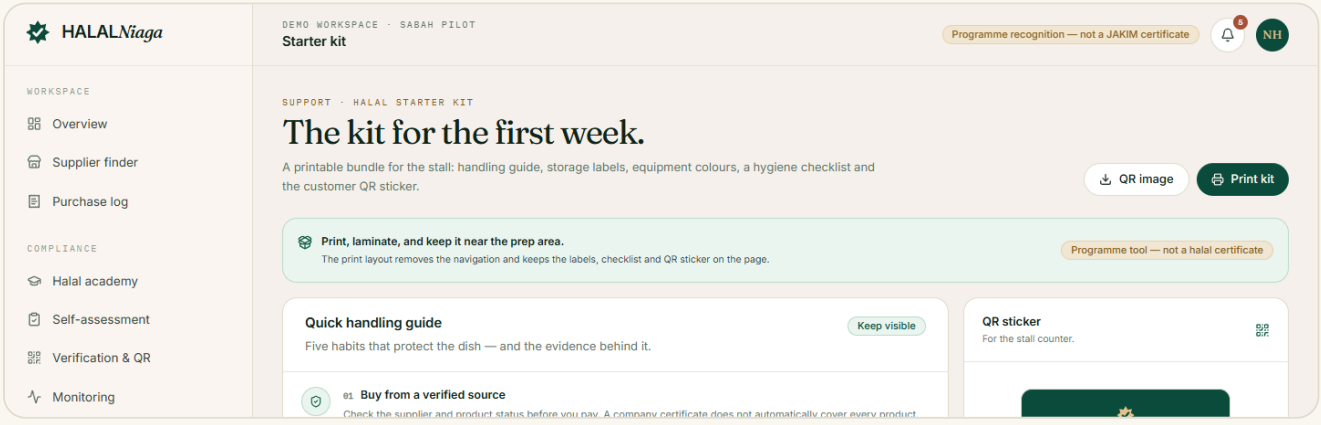
Any email/password opens the demo workspace. No credentials are sent to a server.

Reset

Sidebar → **Reset demo** to restore the original data after a presentation.

5-minute demo flow

1. Open the dashboard: score 68/100, tier Verified, 5 alerts.
2. Supplier Finder → Wet ingredients → Nearest first.
3. Academy → answer a short quiz.
4. Assessment → change one answer, see the score move.
5. Monitoring → record an inspection → tier moves to Monitored.
6. Verification → open the public record / scan the QR.
7. Starter kit → show the labels, checklist and QR sticker.



Starter kit · labels, checklist and QR sticker ready to print.

Presentation tip: start with the public record, then show how each module contributes to the same record. Finish with the starter kit — physical evidence the vendor can take away.

11 · LIMITS & NEXT STEPS

A prototype that is honest about its *limits.*

This prototype demonstrates the service model and user flow. It is not a production compliance system, and it does not make certification decisions.

Prototype limits

- Data is stored in the device browser; it is not synced between devices.
- No backend, real auth, payments or email.
- Supplier certificate status is illustrative, not a live JAKIM lookup.
- Scanner and mentor are simulated.
- Programme inspections are recorded as a simulation.

Next steps

- D1/Workers or Supabase backend for accounts and shared records.
- Supplier onboarding and certificate upload flow.
- JAKIM halal directory integration where authorised access exists.
- Signed QR and an immutable audit trail.
- Physical programme: training, inspection and field recognition.

Programme recognition, not a *halal certificate.*

Malaysian Halal Certification (SPHM) is issued by JAKIM or a recognised state Islamic authority. HALALNiaga supports the compliance journey — it does not replace the official process.

HALALNiaga

halalniaga.pages.dev

UMS student innovation · Sabah pilot 2026

Open the demo
halalniaga.pages.dev · no account required.

Market photographs: Wikimedia Commons — Bahnfrend (CC BY-SA 4.0), CEphoto/Uwe Aranas (CC BY-SA 3.0), Francisco Anzola (CC BY 3.0). Full credits at /credits. Screenshots: HALALNiaga prototype. Logo and interface are original to this project.